AMCC 5160 · LECTURE 07 · 13 OCTOBER 2026

What's New in *AI Motion*

Music, minutes, and machines · Fall 2026

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Four stories from this year

01

The 5-second clip is over

30-second single generations; a real-time minute

Mar – Sep 2026

02

Music becomes the choreographer

A song and a photo give you a minute of dance

Jul 13, 2026

03

Motion leaves the screen

A humanoid learns a K-pop dance from phone video

Jun 10, 2026

04

Sora is gone

What a platform shutdown means for artists

Apr 26 & Sep 24, 2026

Where we've been: six weeks in one slide

01 · Sep 1

Neural networks to AI filmmaking

02 · Sep 8

Generative editing and art direction

03 · Sep 15

Inside the generator: why does it produce this?

04 · Sep 22

AI video generation and art filmmaking

05 · Sep 29

Adaptation, workflows and authorship

06 · Oct 6

Quantization for faster inference (Yaofu)

Tonight pulls these threads together: length (Weeks 3–4), speed (Week 6) and authorship (Week 5).

How many shots is this?

Watch the clip. Count the cuts. Then guess how many times the model was run.

▶ Play: Seedance 2.5 showcase (official page, hero video)

seed.bytedance.com/en/seedance2_5

How top courses treat what's new

Course	What it does with new work	Tonight
Stanford CS348K Fatahalian · Spring 2026	A full lecture on how distillation and caching made autoregressive video real-time; guests from WorldLabs, Roblox, NVIDIA	**Story 1**
CMU 16-726 Jun-Yan Zhu · Spring 2025	Put a same-year fast video paper on the reading list; ends with forensics and data attribution	**Story 2**
MIT 4.562 Nagakura · Fall 2025	Listed Sora as the video tool in the syllabus	**Story 4**
USC IML-460 Holly Willis · Spring 2026	Pairs every tool with the controversy around it	**Each “For the room”**

STORY 1 OF 4

01

The 5-second clip is over

From fragments to scenes, and from rendering to real time

The unit of AI video is now the scene

~5 s

A typical clip, 2024

→

~15 s

Seedance 2.0, early 2026

→

30 s

Seedance 2.5 · Wan 3.0 · Kling 4.0, one pass

→

~60 s

Helios, generated faster than real time

Durations are per single generation, as stated by vendors or reviewers.

Who shipped what

Model	What's new	When
Seedance 2.5 · ByteDance	Built for 30-second storytelling; camera moves and performance blocking	Jul 31, 2026
Wan 3.0 · Alibaba	30-second clips; beta only, no open weights	Public beta, Aug 2026
Kling 4.0 · Kuaishou	Announced 3–30 s, up to 4K; Flash tier 20 s at 720p	Flash Sep 28; full model October
Helios · Peking Univ. + ByteDance	14B model; ~60 s at 19.5 FPS on one H100; open weights	Mar 2026

Seedance 2.5 official page ↗

seed.bytedance.com/en/seedance2_5

Why five seconds was the limit

6× the frames

At 24 fps, 5 s is 120 frames. 30 s is 720 frames.

~36× the attention

Full attention compares every token with every other one, so six times the tokens means roughly 36 times the work.

Errors pile up

The longer the video, the more chances for small mistakes to grow. That is drift, and we'll come back to it.

The 36× is a rule of thumb for full attention; real systems use many tricks to beat it.

Two ways to make a long video

All at once

- Denoise every frame of the clip together
- Strong consistency inside the clip
- Cost grows fast with length; you wait for the whole thing

Forward in time (autoregressive)

- Write the next chunk from the chunks before it
- Can stream and run in real time
- Must fight drift: errors carry forward

Helios is the second kind. Language models write text the same way: one piece after another.

Helios: a minute of video, in real time



Frame from the official Helios feature video

- 14B autoregressive model: writes video forward, chunk by chunk
- ~60 s (1,440 frames) per generation
- 19.5 fps on one H100: close to 24 fps playback
- Open weights

▶ Play: Helios feature video (YouTube)

www.youtube.com/watch?v=vd_AgHtOUFQ

▶ Minute-long demos (project page)

pku-yuangroup.github.io/Helios-Page

Try it: Hugging Face demo ↗

huggingface.co/spaces/BestWishYsh/Helios-14B-RealTime

How Helios gets its speed

Remember less, smartly

It compresses the history it carries forward, so each new chunk doesn't re-read the whole past.

Cheaper than its size

The team reports per-step cost comparable to a 1.3B model, despite having 14B parameters.

Fewer steps

A distilled version uses 3 denoising steps instead of dozens (Week 6's theme: same model, less work).

Lean to train

Up to four 14B models fit in 80 GB of GPU memory during training, without special parallelism frameworks.

▶ Speed demo (project page, “Inference Speed”)

pku-yuangroup.github.io/Helios-Page

Why long video falls apart: drift

Second 1

The character looks right

→

Second 20

Small errors feed the next chunk

→

Second 60

Face, clothes and room have changed

Helios's fix: train on deliberately drifted video, so the model learns to recover instead of patching drift at playback time.

Test it yourself: does she come back the same?

On the Helios project page, find “Verify Consistency”: a subject leaves the frame and returns. Then try “Interactive-Video Showcases”, where the prompt changes while the video keeps going.

▶ Play: Helios consistency and interactive demos

pku-yuangroup.github.io/Helios-Page

Real time changes what video is

From watching to playing

If frames arrive faster than you watch them, you can steer the video as it plays.

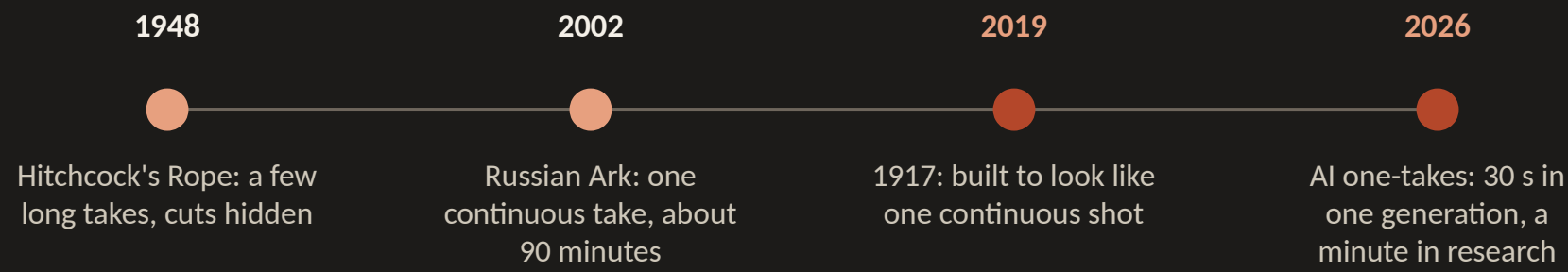
Stanford taught this in May

CS348K spent a lecture on how distillation and caching made autoregressive video real-time.

Coming in Week 10

Interactive world models and explorable art (Nov 3) start from exactly this idea.

The long take is an old dream



What 30 seconds unlocks: blocking

Performance blocking

Where actors stand and move across a whole scene, not one gesture.

Camera movement

A move with a start, a middle and an end, inside one take.

Green-screen and white-model control

Bring in your own blocked-out scene or plates to steer the result.

Extend twice

Continue a 30-second generation up to two more times.

▶ **Seedance 2.5 official page (feature videos)**

seed.bytedance.com/en/seedance2_5

FOR THE ROOM

If one generation is a whole 30-second scene, what is the director's job now?

And what did the editor just lose?

STORY 2 OF 4

02

Music becomes the choreographer

A song, one photo, a minute of dance

AI dance has a short, fast history

2019

Everybody Dance Now (Berkeley): copy a dancer's pose onto you

2021

AI Choreographer (Google): music to 3D dance motion

2023

EDGE (Stanford): editable dance from music

2026

Wan-Dancer: music + photo to finished video, a minute long

Wan-Dancer: song in, dance out

A music track

One photo of the dancer

A style word

→

A 1-minute+ dance video, 720p / 30 fps, on the beat

- Released Jul 13, 2026 by Tongyi Lab (Alibaba) as open weights, Apache 2.0
- Five styles: Chinese classical, K-pop, street, tap, Latin

▶ Play: Wan-Dancer demos (project page)

humanaigc.github.io/wan-dancer-project

Paper (arXiv 2607.09581) ↗

arxiv.org/abs/2607.09581

Skeleton then, pixels now

Before: music → skeleton

- Output is 3D joint motion
- You still need a character, costume, lighting and a renderer
- Great for games and animation pipelines

Now: music + photo → video

- Output is the finished image
- Look, light and camera come from the model
- Fast to try, harder to edit afterwards

The trade-off: control over every joint, or a finished shot in one step.

One dancer, many looks; one song, many dancers



Reference images from the Wan-Dancer project page

- Swap the photo, keep the song: the same dance, a different dancer
- Swap the song, keep the photo: the same dancer, a new dance
- Earlier music-to-dance models fell apart after about 20 seconds

▶ Play: compare with earlier models (§6)

humanaigc.github.io/wan-dancer-project

The trick: plan the song, then the steps

Global level: the whole song

Intro	Verse	Build	Chorus	Outro

↓ plans

Local level: the moves

Renders each passage's steps, beat by beat

This is how a choreographer works: structure first, steps second.

What the model has to hear

Beat

Where the accents fall: a step, a hit, a tap must land on them.

Tempo

How fast the beats come: the same move looks different at 90 and 140 bpm.

Phrases and sections

Eight-counts, verses, choruses: where a movement idea starts and ends.

Energy

Quiet to loud, sparse to dense: when to hold still and when to explode.

Five styles, five tests

Chinese classical

Flow: does the movement stay continuous through the phrase?

K-pop

Precision: are the hits sharp and exactly on the beat?

Street

Groove: does the body bounce with the rhythm, not just pose?

Tap

Sound: do the feet land where the taps are heard?

Latin

Hips and rhythm: is the movement driven from the core?

Your verdict

Which style does the model handle worst, and why?

▶ Play: all five styles (project page, section 1)

humanaigc.github.io/wan-dancer-project

Machines in choreography are not new

Merce Cunningham · chance

Used chance procedures, such as tossing coins, to order movements and decide structure.

Cunningham · Trackers, 1991

The first dance he made partly with LifeForms, an early computer program for figures in motion.

Wayne McGregor · Living Archive, 2019

An AI tool trained on his 25-year archive suggests the next phrase; dancers translate it through their bodies.

Living Archive (Studio Wayne McGregor) ↗

waynemcgregor.com/productions/living-archive

Limits to keep in mind

Hardware

Open weights, but its release materials point to datacenter-class GPUs, not a laptop.

One dancer, one photo

The look comes from a single image; everything unseen in it is invented.

Whose moves?

Viewers cannot check whose dances the training videos contained.

Music has rights

The song that drives the dance is someone's work; credit it and check its licence.

Pitch a one-take

5 MINUTES · IN PAIRS

1

One song: name it, and say why

2

One photo: who or what dances?

3

One sentence: what happens in the 30 seconds?

Three pairs read theirs aloud. Keep your pitch: it could become Assignment 2.

FOR THE ROOM

Who is the choreographer here?

The musician? The model? The thousands of dancers in its training videos? The person who picked the song?

STORY 3 OF 4

03

Motion leaves the screen

A phone video becomes a robot's dance

Robots have danced before

2000s

HRP-2 learns a Japanese folk dance from motion capture

Dec 2020

Boston Dynamics “Do You Love Me?”: hand-authored routine

Jun 2026

AI Sapiens learns a K-pop challenge from phone video

A humanoid learns a K-pop challenge from phone video



ROBOTIS AI Sapiens dancing the CORTIS “REDRED” challenge (ROBOTIS video thumbnail)

- ROBOTIS (Korea), open-source humanoid AI Sapiens, June 2026
- 1.3 m tall, 34 kg, 23 joints
- Learned from ordinary 2D video: no motion-capture studio

▶ Play: ROBOTIS dance video (YouTube)

www.youtube.com/watch?v=NeL2BmgcFJs

News story (Interesting Engineering) ↗

interestingengineering.com/ai-robotics/korean-humanoid-r…

Four steps from phone to robot

1

Film

A dancer, on a phone

→

2

Lift

Recover 3D joint motion from the 2D video

→

3

Practise

Reinforcement learning in simulation fixes balance and timing

→

4

Transfer

Move the learned skill to the real robot

Every step is an idea from this course: pose from video, simulation, and a model of the world.

Why not just copy the pose?

Different body

1.3 m, 34 kg, 23 joints: limbs and proportions don't match the dancer's.

Speed limits

Motors can't accelerate like muscles. A 2025 study copying a dancer's arms found the robot couldn't keep the rhythm.

Balance

A human shifts weight without thinking. Copy the pose exactly and the robot falls over.

Study: robot arms copying a dancer (2025) ↗

doaj.org/article/b55086ceb9bf4dd3b6b2893ef09cb7db

The sim-to-real gap

Practice in a copy of the world

The robot's twin trains in NVIDIA Isaac Sim, falling thousands of times at no cost.

But the copy is imperfect

Friction, motor delay and flex differ between simulation and the real floor.

Close the gap in hardware

ROBOTIS designed its DYNAMIXEL-Q actuators so simulated joints behave more like real ones.

A course idea in disguise

Simulation is a world model. We'll meet video world models in Week 10.

Why dance challenges are perfect training data

Short

Fifteen to thirty seconds: a complete, learnable unit.

Repeated

Thousands of people film the same routine: many examples of one choreography.

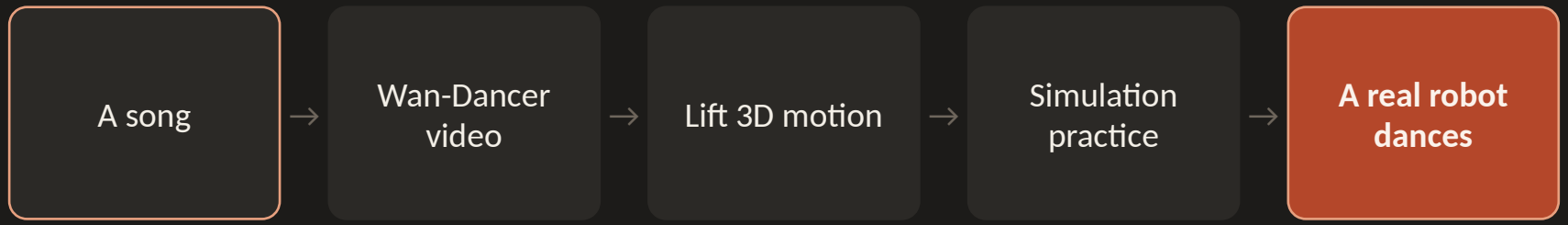
On the beat

The music marks exactly when each move should happen.

Full body, facing the camera

Made to be watched and copied, which also makes them easy to read.

A thought experiment



Speculation: as far as public reports go, nobody has shown this end to end. But every piece now exists.

Video is becoming training data for bodies

For robots

Human videos teach skills without motion capture: dance today, chores tomorrow.

For world models

Video generators that predict what happens next can act as simulators (Week 10).

For artists

Your choreography video may have two audiences: people, and machines that learn from it.

FOR THE ROOM

If a generated video can teach a real body to move, is that video still “fake”?

What is a dance video for, once machines learn from it?

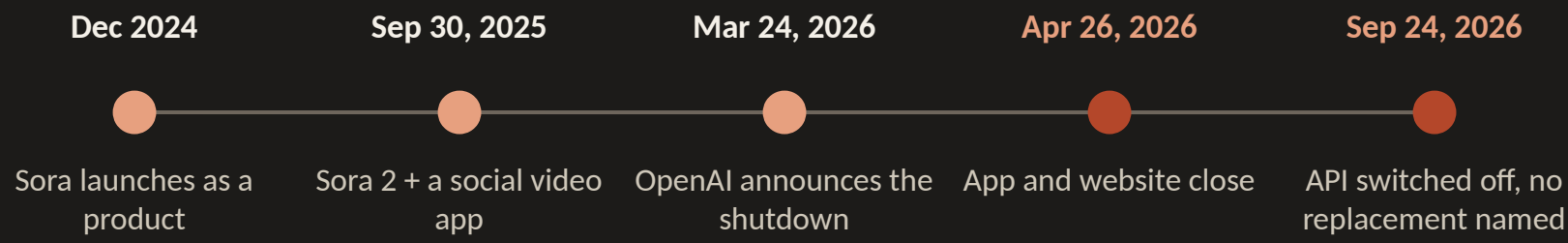
STORY 4 OF 4

04

Sora is gone

When your tool disappears

Eighteen months as a product



OpenAI: What to know about the Sora discontinuation ↗

help.openai.com/en/articles/20001152-what-to-know-about-...

Open or closed, this year

Model	Status	Can you still run it in 5 years?
Helios	Open weights	Yes, if you keep a copy
Wan-Dancer · Wan 2.2	Open weights (Apache 2.0)	Yes, if you keep a copy
Seedance 2.5 · Wan 3.0 · Kling 4.0	Closed; app or API only	Only if the company keeps it running
Sora 2	Discontinued	No

This has happened before: Flash

Dec 31, 2020

Adobe ended support for Flash Player. Two decades of web animation and games stopped playing in browsers.

Rescue by emulation

The Internet Archive keeps thousands of Flash works playable with an emulator, Ruffle.

Art institutions

Rhizome's ArtBase has preserved born-digital art since 1999. Preservation is a practice, not an accident.

Last fall, a top syllabus was built on Sora

MIT's Fall 2025 motion-graphics course listed Sora as its video tool. Some of last year's student projects depend on a tool that no longer exists.

Keep your sources, not just your outputs

- [x] Prompts, word for word
- [x] Seeds and settings
- [x] Reference images, clips and songs (with credit)
- [x] Raw generations, before editing
- [x] Edit project files
- [x] Model name and version, and the date

Why open weights matter

Open-weight models (Wan 2.2, Helios, Wan-Dancer) can still be run years from now. Closed apps can't.

A project folder that survives

README.md

Model, version, date, settings, and what you were trying to do.

prompts/

Every prompt you kept, word for word, with its seed.

refs/

Images, clips and songs you used, with credits and licences.

raw/

Unedited generations, including the failures worth remembering.

edit/

The editing project and the exported master.

Start today

Use this structure for Assignment 2.

FOR THE ROOM

What does an AI artwork need in order to outlive the tool that made it?

Think of a film, a painting, a game. What survives, and why?

Four stories, one line each

01 · Minutes

The unit of AI video moved from the shot to the scene, and toward real time.

02 · Music

A song and a photo now give you a minute of dance, planned like a choreographer would.

03 · Machines

Phone video can teach a real robot to dance; video now has machine audiences.

04 · Loss

Tools can vanish. Keep your sources; prefer what you can still run.

DEBATE · 10 MINUTES

“A generated dance that a robot can learn is a real dance.”

Left side of the room argues for; right side against

Two speakers per side, one minute each

Then the whole room votes by show of hands

Debate starters

For: it is a real dance

- Choreography is a structure of movement in time; the model plans one
- A real body performs it, with real risk of falling
- Dance has always borrowed from tools: chance, software, video

Against: it is not

- No one intended it; a model averaged other people's dances
- The robot executes, it doesn't express
- Without a dancer's choice, it's motion, not dance

Use one example from tonight in your minute.

This week

READING 1 · TECHNIQUE

Helios: real-time long video generation

READING 2 · ART + TECH

Wan-Dancer project page

READING 3 · CULTURE

What happened to Sora?

Quiz 07 · two technique, two art, one reflection · hyang.org/amcc5160/quiz-07

Assignment 2 is due Oct 31 · Next week, Lecture 08: native audio-video, performance and real-time characters

All demo links in one place

STORY 1

- ▶ Seedance 2.5 official page
- ▶ Helios feature video (YouTube)
- ▶ Helios project page

STORY 2

- ▶ Wan-Dancer project page
- ▶ Wan-Dancer weights (Hugging Face)
- ▶ Living Archive (Wayne McGregor)

STORY 3

- ▶ ROBOTIS: Learning from Videos (YouTube)
- ▶ AI Sapiens documentation
- ▶ Study: robot arms copying a dancer

STORY 4

- ▶ OpenAI: Sora discontinuation

Music, minutes, and machines

Motion got longer, found a new choreographer, left the screen, and lost a tool. Keep your sources.

Sources: Seedance 2.5 (ByteDance Seed); Helios (Peking University, ByteDance, arXiv 2603.04379); Wan-Dancer (Tongyi Lab, arXiv 2607.09581); ROBOTIS AI Sapiens; OpenAI Help Center; reviews by Pinggy, Atlas Cloud and Unifically for Wan 3.0 and Kling 4.0 specs. Images: Helios feature video, Wan-Dancer project page, ROBOTIS video thumbnail, used for teaching.